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[Personal Portfolio Website](#)

Artificial Intelligence and Computer Vision Engineer

PROFILE SUMMARY

Computer Vision and AI Engineer with **5+ years of industrial experience** — from shipping ML models to 300K+ fleet customers at Motive (**Forbes Cloud 100 company**, VTTI-ranked #1 AI dashcam) to prototyping AR surgical guidance at Cochlear. **MSc in Computer Vision** and Robotics (Université Bourgogne Europe / UdG, **Erasmus Scholar**). Specialized in Deep Learning, 3D Computer Vision, SLAM, ROS, LLMs and Agentic AI systems — across the full stack from model development to production deployment on cloud and edge.

EDUCATION

UNIVERSITÉ BOURGOGNE EUROPE

France

M.Sc Computer Vision and Robotics ([VIBOT](#))

Sep 2024 - June 2026

- **Notable Courses:** Computer Vision, Advanced Machine Learning, Autonomous Robotics, 3D Perception & Sensor Fusion, Reinforcement Learning, Medical Imaging Analysis, Signal Processing
- **Erasmus+ Exchange:** *MS Intelligent Robotics Systems* at [University of Girona, Spain](#)
- **Thesis:** Augmented Reality (AR) Assistive Surgery for Cochlear Implants at [Cochlear Ltd](#)

GIK INSTITUTE OF ENGINEERING SCIENCES AND TECHNOLOGY (GIKI)

Pakistan

B.S Engineering Sciences (Photonics and Optoelectronics)

2016 - 2020

- **Courses:** Robotic Vision, Applied Artificial Intelligence, Data Structures, Object Oriented Programming (OOP)
- **Accreditation:** [Pakistan Engineering Council \(PEC\)](#) and [Washington Accord](#)
- **Extracurriculars:** President at [SPIE GIKI Branch](#) & Team Captain at GIKI Robotics Team Technobolt
- **Awards:** Best Final Year Project Awards at GIKI; secured IGNITE's National ICT Grant by Government

WORK EXPERIENCE

COCHLEAR

France

Computer Vision Engineer, R&D

Feb 2026 – Present

- Developing a **first-of-its-kind AR surgical guidance system** at one of the world's biggest medical tech companies — directly enabling the **3× surgeon scale** required to meet global CI demand by 2030
- Built **Visual-Inertial SLAM** (deep learning + geometric approaches, RGB-D + IMU) for real-time dense **3D Reconstruction**; automatic **3D CT-to-scene registration**; neural segmentation for **live AR overlay** of anatomical structures — evaluated ORB-SLAM3 + TSDF fusion and MAST3R-SLAM with loop closure for surgical precision grade drift correction. Using FoundationPose for 6DoF Pose Track
- Replaced all prior infeasible prototypes; **20 times cost reduction** through algorithm-first design

BITTECH SOLUTIONS

Remote

Senior AI Engineer

Sep 2025 – Feb 2026

- Led end-to-end AI delivery across **10+ enterprise clients** as **solutions architect** and **team lead** — defined system designs, aligned clients on technical scope, deployment on cloud, and **200% increase** in closing
- Built production **RAG pipelines** (LangChain, LlamaIndex, Ollama) with citation-accurate retrieval
- Developed **Agentic AI** and **workflow automation** systems (LangGraph, MCP, n8n)

MOTIVE

Remote, USA 

Software Engineer, Machine Learning (R&D Department)

June 2021 – September 2024

- 3+ years building production **AI dashcam** for driver safety, at a **Forbes Cloud 100** company — VTTI-ranked **#1 AI Dashcam**, deployed to **300K+ commercial fleets** across North America.
- **Computer Vision Models:** Trained and deployed image classification and object detection models like **EfficientDet** & **YoloX** with Knowledge Distillation; implemented **Stateless LSTM/ConvLSTM** on frame-level object detector embeddings for real-time **Activity Recognition**. Trained **Diffusion Models** for synthetic data
- Developed heuristics based filters to the **post-processing engine** that aided in suppressing potential false positives, resulting in **10% increase in feature precision**. FFT on telematics data for drowsiness detection
- **Data Engineering & MLOps:** Automated **AWS** pipelines (EC2, S3, Lambda) for large-scale extraction; **SQL Queries** and dashboards tracking production failure modes — iterative model improvements and field analysis drove **35% cumulative accuracy gain**, contributing to **22% reduction in client accident rates**

FACILITY OF ADVANCED ROBOTICS AND INTELLIGENT SYSTEMS (109-AED)

Pakistan 

R&D Engineer - Robotics and Perception

August 2020 – June 2021

- Researched, prototyped, and deployed **4+ perception and robotics systems** — autonomous ground vehicles, aerial drones, and training simulators — spanning image processing pipelines, deep learning inference, and deployed on embedded edge processors like **Nvidia Jetson TX2**
- Implemented **Extended Kalman Filter** sensor fusion across IMU, camera, and ranging sensors; full hardware integration via I2C, SPI, UART, RS232.

SKILLS

- **Programming & Systems:** Python · C++ · CUDA · SQL · Bash / Linux
- **Machine Learning & Deep Learning:** PyTorch · TensorFlow · CNNs · Vision Transformers (ViT, DINO) · Object Detection (YOLOx, EfficientDet) · Image Segmentation (SAM) · LSTM / ConvLSTM · Knowledge Distillation · Model Quantization & Optimization
- **3D Computer Vision & SLAM:** Visual-Inertial Odometry · ORB-SLAM3 · MAST3R-SLAM · Structure-from-Motion · Stereo Vision · Depth Estimation · ICP Registration · Open3D / PCL · RealSense
- **Robotics:** ROS / ROS2 · Extended Kalman Filter · Sensor Fusion (IMU, LiDAR, Camera) · Reinforcement Learning (PPO, SAC) · Nvidia Jetson TX2 · Raspberry Pi · Embedded Systems
- **LLMs, RAG & Agentic AI:** LangChain · LangGraph · LlamaIndex · RAG Pipelines · Agentic AI · Vector Databases (ChromaDB, FAISS) · BGE-M3 · Ollama · VLMs · Whisper · Multimodal AI · MCP
- **MLOps & Cloud:** AWS (EC2, S3, Lambda, SageMaker) · Docker · n8n · Edge Deployment · CI/CD · Git

PROJECTS

- **Visual Inertial SLAM** for 3D Reconstruction (OrbSLAM3+TSDF Fusion, MAST3R-SLAM)
- 3D Reconstruction using Incremental **Structure from Motion (SfM)** - [\[GitHub Link\]](#)
- **RAG-based self-hosted LLM** for Customer Support for Enterprise
- Multi-App ETL Autonomous **AI Workspace Orchestrator (n8n)** - [\[Notion Link\]](#)
- **Bio-Inspired Underwater Robot** with Undulating Fin Based Locomotion
- **BigGAN-lite** for Synthetic Face Generation
- **Satellite Image Analysis** for Vegetation and Detections - [\[GitHub Link\]](#)
- Multi-lingual Video Dubbing and GAN based Lip-sync - [\[GitHub Link\]](#) / [\[Notion Link\]](#)
- Obstacle Avoidance and Path Planning of a TurtleBot using Lidar and **Reinforcement Learning**
- **Activity Recognition** using YOLOv5 Embeddings fed into LSTM for dynamic length spatio-temporal context
- **Job Application Agent** that searches, generated a relevant CV, and auto applies to various job platforms